Practical 9

Problem definition:

You are a Big Data Analyst at TrendPulse Social Media Analytics, a company that monitors social media activity across multiple platforms such as Twitter, Instagram, Facebook, and LinkedIn. Every day, millions of users post content containing hashtags related to current events, technology, sports, entertainment, politics, and business. The marketing team wants to identify: most trending hashtags, popular discussion topics, user interests, and emerging market trends. Traditional analytics tools cannot efficiently process the massive volume of social media data generated every day.

Objectives:

1. To create a Spark Session for processing social media data.
2. To load and validate a social media hashtag dataset using Spark DataFrames.
3. To group identical hashtags and calculate their frequency.
4. To identify the top 3 trending hashtags based on their occurrence.
5. To compare hashtag trends across multiple days.
6. To analyze hashtag popularity across different regions.
7. To classify hashtags into categories such as Technology, Sports, and Entertainment.
8. To generate visualizations for hashtag trends and category popularity.
9. To interpret user interests and emerging trends.
10. To provide business and digital marketing recommendations based on the analysis.

Observation / Output:

Task 1: Create Spark Session.

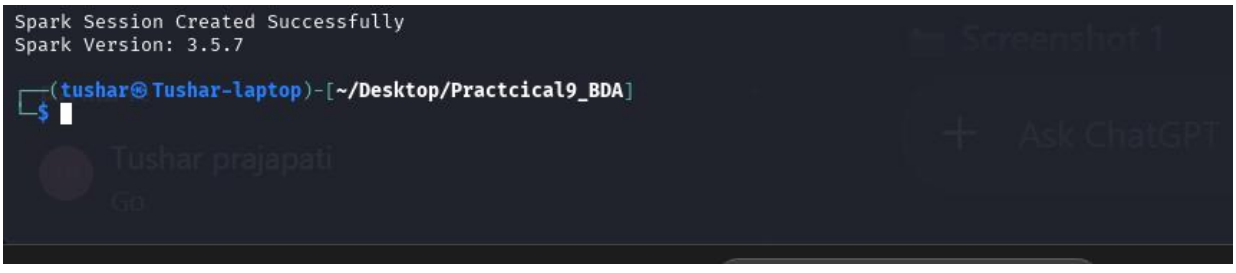
```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

print("Spark Session Created Successfully")
print("Spark Version:", spark.version)

spark.stop()
```



Task 2: Load Social Media Hashtag Dataset

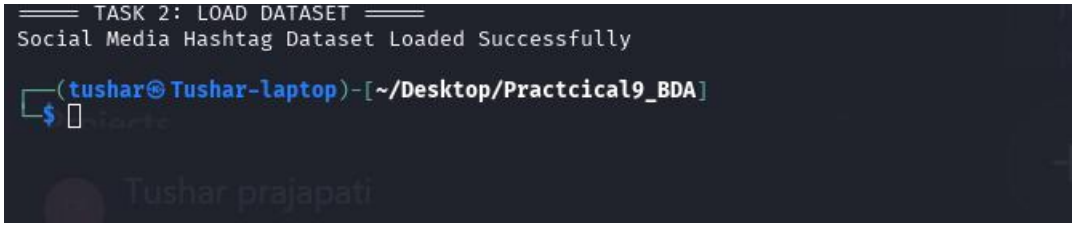
```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("SocialMediaHashtagAnalysis") \
    .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
    "social_media_hashtags.csv",
    header=True,
    inferSchema=True
)
print("Social Media Hashtag Dataset Loaded Successfully")

spark.stop()
```



Task 3: Display Hashtag Records

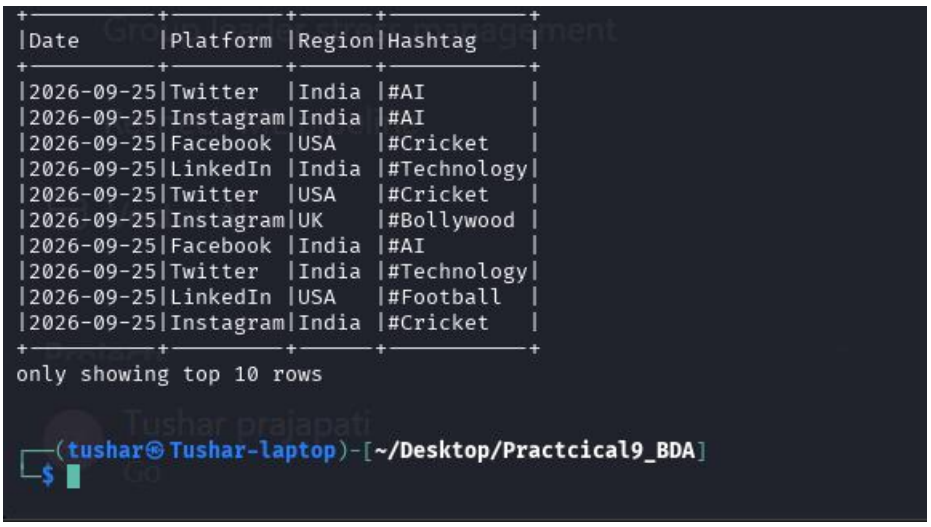
```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("SocialMediaHashtagAnalysis") \
    .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
    "social_media_hashtags.csv",
    header=True,
    inferSchema=True
)
df.show(10, truncate=False)

spark.stop()
```



Task 4: Validate Dataset Structure

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

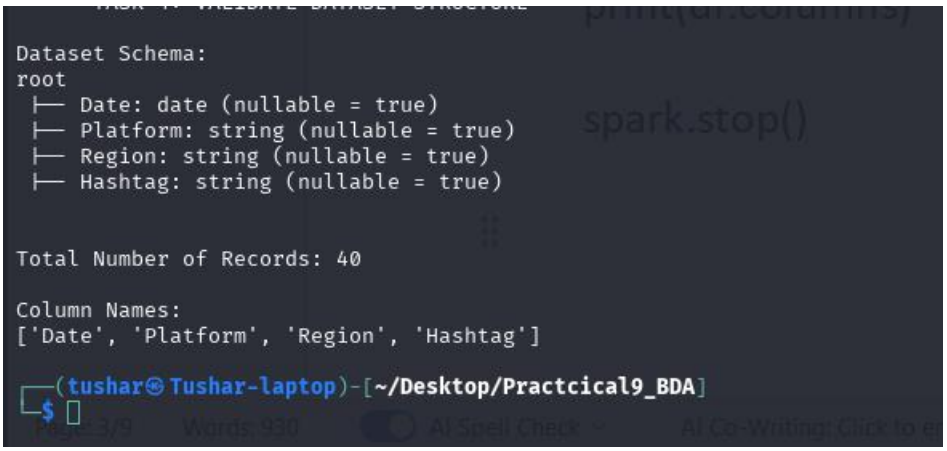
df = spark.read.csv(
   "social_media_hashtags.csv",
   header=True,
   inferSchema=True
)

print("\nDataset Schema:")
df.printSchema()

print("\nTotal Number of Records:", df.count())

print("\nColumn Names:")
print(df.columns)

spark.stop()
```



Task 5 : Group Identical Hashtags

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

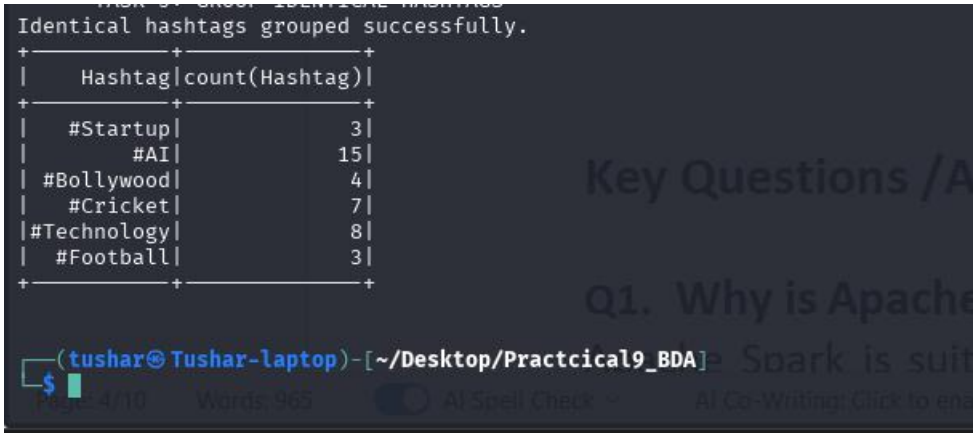
spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
   "social_media_hashtags.csv",
   header=True,
   inferSchema=True
)
grouped_hashtags = df.groupBy("Hashtag")

print("Identical hashtags grouped successfully.")

grouped_hashtags.agg({"Hashtag": "count"}).show()

spark.stop()
```



Task 6 — Count Hashtag Occurrences

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import count
```

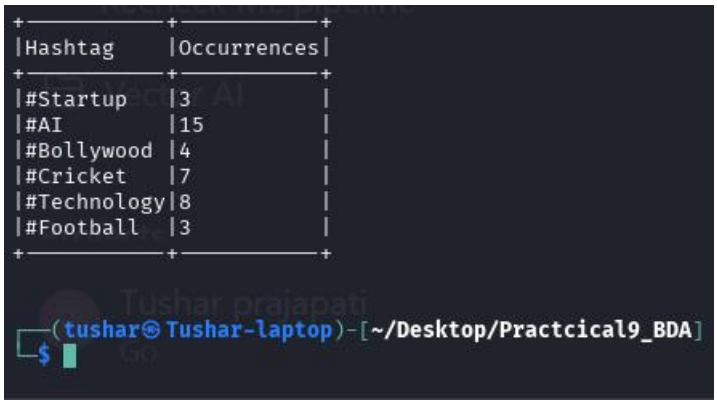
```
spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
   "social_media_hashtags.csv",
   header=True,
   inferSchema=True
)
hashtag_counts = df.groupBy("Hashtag") \
   .agg(count("*").alias("Occurrences"))

hashtag_counts.show(truncate=False)

spark.stop()
```



Task 7 — Sort Hashtags Based on Popularity

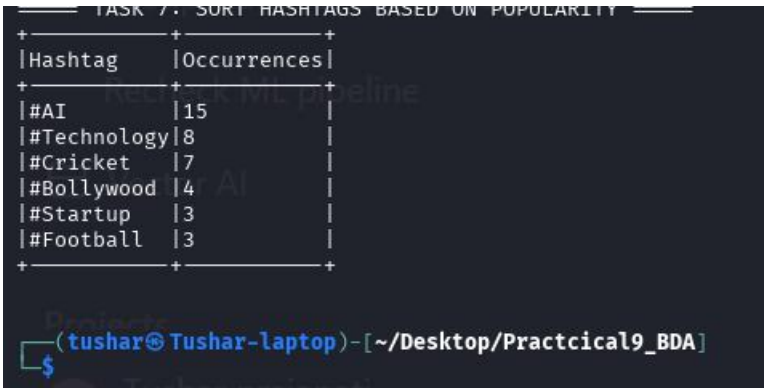
```
from pyspark.sql import SparkSession
from pyspark.sql.functions import count, desc
spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")
df = spark.read.csv(
   "social_media_hashtags.csv",
   header=True,
   inferSchema=True
)
hashtag_counts = df.groupBy("Hashtag") \
   .agg(count("*").alias("Occurrences"))

sorted_hashtags = hashtag_counts.orderBy(
   desc("Occurrences")
)

sorted_hashtags.show(truncate=False)

spark.stop()
```



Task 8 - Identify Top Trending Hashtags

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import count, desc

spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
   .getOrCreate()

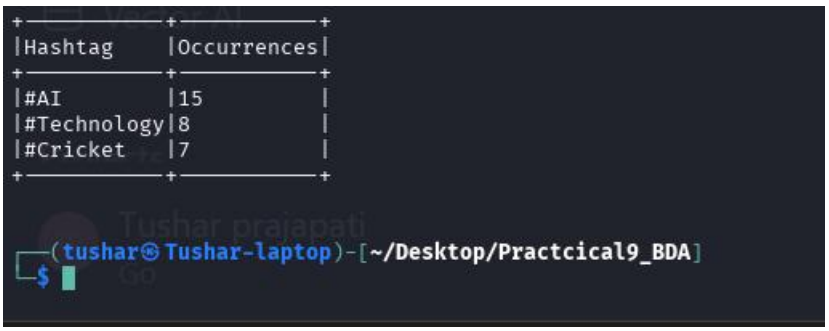
spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
   "social_media_hashtags.csv",
   header=True,
   inferSchema=True
)
hashtag_counts = df.groupBy("Hashtag") \
   .agg(count("*").alias("Occurrences"))

top_trending = hashtag_counts.orderBy(
   desc("Occurrences")
).limit(3)

top_trending.show(truncate=False)

spark.stop()
```



Task 9 - Generate Trend Analysis Report

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import count, desc

spark = SparkSession.builder \
   .appName("SocialMediaHashtagAnalysis") \
```

```
    .getOrCreate()

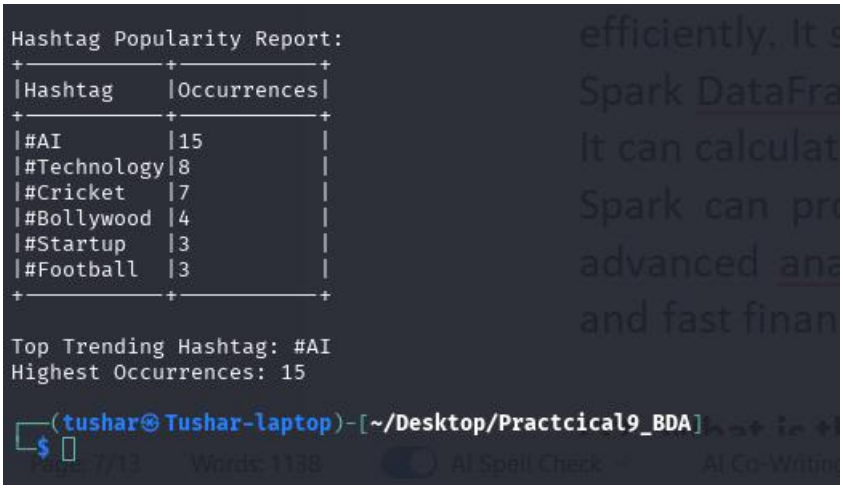
spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
    "social_media_hashtags.csv",
    header=True,
    inferSchema=True
)
hashtag_counts = df.groupBy("Hashtag") \
    .agg(count("*").alias("Occurrences")) \
    .orderBy(desc("Occurrences"))

print("\nHashtag Popularity Report:")
hashtag_counts.show(truncate=False)

print("Top Trending Hashtag:", hashtag_counts.first()["Hashtag"])
print("Highest Occurrences:", hashtag_counts.first()["Occurrences"])

spark.stop()
```



Task 10 - Interpret User Interests

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import count, desc

spark = SparkSession.builder \
    .appName("SocialMediaHashtagAnalysis") \
    .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")

df = spark.read.csv(
    "social_media_hashtags.csv",
    header=True,
    inferSchema=True
)
hashtag_counts = df.groupBy("Hashtag") \
    .agg(count("*").alias("Occurrences")) \
    .orderBy(desc("Occurrences"))
```

```
print("\nUser Interest Analysis:")
hashtag_counts.show(truncate=False)

print("\nInterpretation:")
print("#AI shows the highest user interest with 15 occurrences.")
print("#Technology is the second most popular topic with 8 occurrences.")
print("#Cricket is also a major area of user interest with 7 occurrences.")
print("Entertainment and other topics have comparatively lower popularity.")

spark.stop()
```

```
User Interest Analysis:
+----------+-----------+
|Hashtag   |Occurrences|
+----------+-----------+
|#AI       |15         |
|#Technology|8         |
|#Cricket  |7          |
|#Bollywood|4          |
|#Startup  |3          |
|#Football |3          |
+----------+-----------+


Interpretation:
#AI shows the highest user interest with 15 occurrences.
#Technology is the second most popular topic with 8 occurrences.
#Cricket is also a major area of user interest with 7 occurrences.
Entertainment and other topics have comparatively lower popularity.

┌──(tushar㉿Tushar-laptop)-[~/Desktop/Practcical9_BDA]
└─$
```

Task 11 - Suggest Business Recommendations

- **Focus on AI and Technology trends:** Since `#AI` and `#Technology` have the highest occurrences, businesses can create technology-focused content and campaigns around these topics.

- **Use Sports trends:** The popularity of `#Cricket` indicates an opportunity for businesses to create sports-related promotional campaigns and audience engagement activities.

- **Use trending hashtags in marketing campaigns:** Businesses can include popular hashtags in social media posts to improve visibility and reach.

- **Monitor changing trends:** Regular hashtag analysis can help businesses identify emerging topics and adjust their marketing strategies accordingly.

- **Target content according to user interests:** The identified interests can help businesses design relevant content, promotions and campaigns for their target audience.

Key Questions /Analysis:

Q1. Why is Apache Spark suitable for social media analytics?

Apache Spark is suitable for social media analytics because it can process large amounts of data quickly. It supports distributed processing across multiple systems. Spark DataFrames make data loading, filtering and aggregation easier. It is useful for analyzing large-scale social

media data and identifying trends efficiently.

Q2. What is the significance of hashtag frequency?
Hashtag frequency shows how often a particular hashtag is used. A higher frequency indicates greater popularity or user interest in that topic. It helps identify trending topics and popular discussions. Businesses can use this information to understand audience interests.

Q3. How can trend analysis support digital marketing campaigns?
Trend analysis helps businesses identify topics that are currently popular among users. Marketers can create relevant content based on these trending topics. Popular hashtags can be used to improve social media reach and engagement. This helps businesses make their marketing campaigns more effective.

Q4. Why do businesses monitor social media trends?

Businesses monitor social media trends to understand changing customer interests. It helps them identify emerging topics and opportunities in the market. Trending information can guide content creation and promotional campaigns. It also helps businesses stay competitive and respond to customers effectively.

Q5. How can real-time analytics improve decision making?

Real-time analytics provides businesses with the latest information about user activity and trends. It allows quick identification of changes in customer interests or market conditions. Businesses can respond immediately by modifying their content or marketing strategies. This supports faster and more informed decision making.

Supplementary Problems:

Identify top 3 trending hashtags. Compare trends across multiple days. Analyze hashtag popularity by region. Create hashtag categories (Technology, Sports, Entertainment). Generate trend visualization using charts.

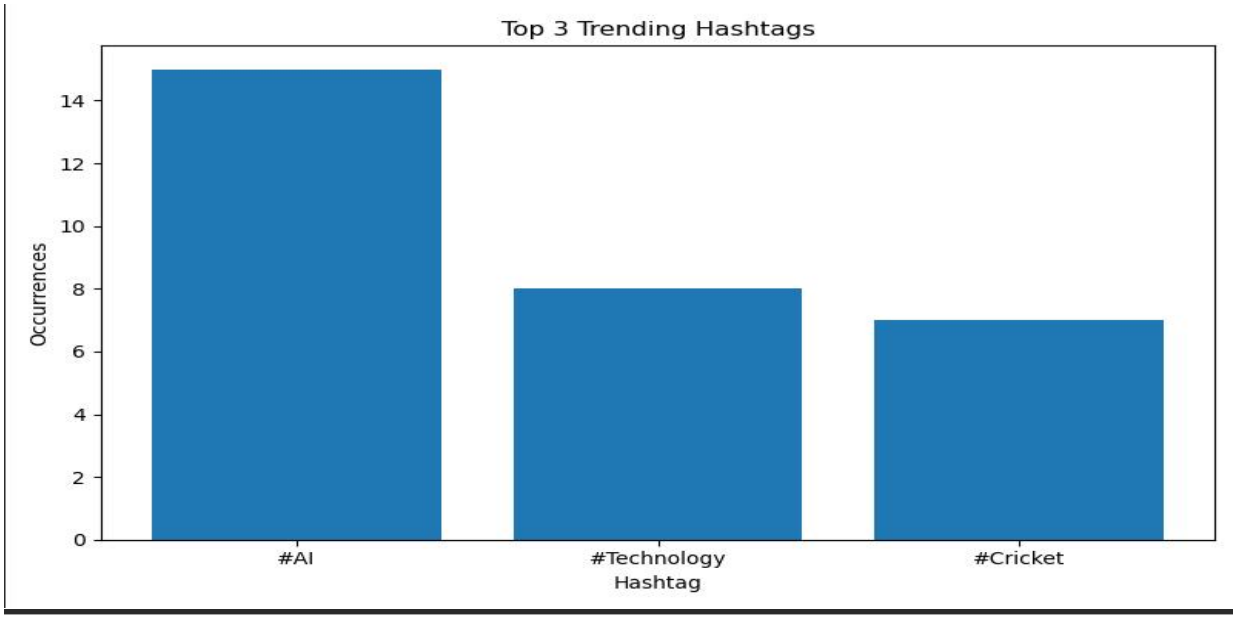
```
+-----------+-----------+
|Hashtag    |Occurrences|
+-----------+-----------+
|#AI        |15         |
|#Technology|8          |
|#Cricket   |7          |
+-----------+-----------+

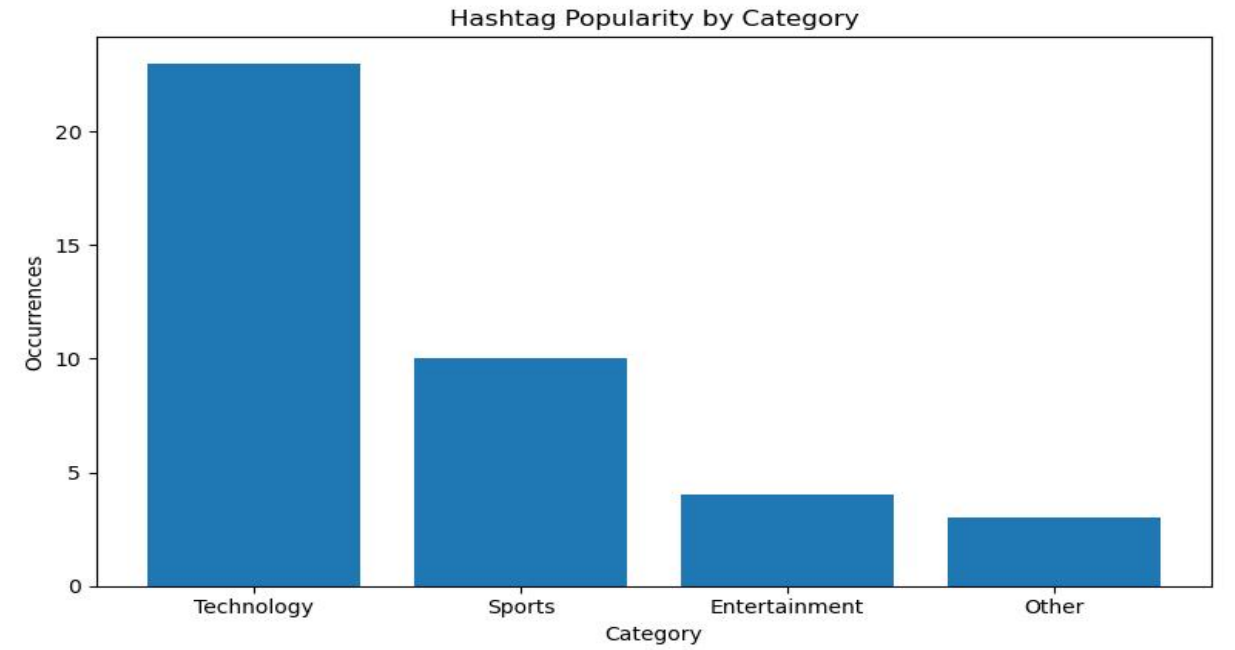

===== 2. TREND COMPARISON ACROSS MULTIPLE DAYS =====
+----------+-----------+-----------+
|Date      |Hashtag    |Occurrences|
+----------+-----------+-----------+
|2026-09-02|#AI        |1          |
|2026-09-25|#AI        |3          |
|2026-09-25|#Cricket   |3          |
|2026-09-25|#Technology|2          |
|2026-09-25|#Football  |1          |
|2026-09-25|#Bollywood |1          |
|2026-09-26|#AI        |4          |
|2026-09-26|#Technology|2          |
|2026-09-26|#Cricket   |1          |
|2026-09-26|#Bollywood |1          |
|2026-09-26|#Startup   |1          |
|2026-09-27|#AI        |3          |
|2026-09-27|#Technology|2          |
|2026-09-27|#Cricket   |2          |
|2026-09-27|#Startup   |1          |
|2026-09-27|#Football  |1          |
|2026-09-27|#Bollywood |1          |
|2026-09-28|#AI        |4          |
|2026-09-28|#Technology|2          |
|2026-09-28|#Bollywood |1          |
+----------+-----------+-----------+
only showing top 20 rows

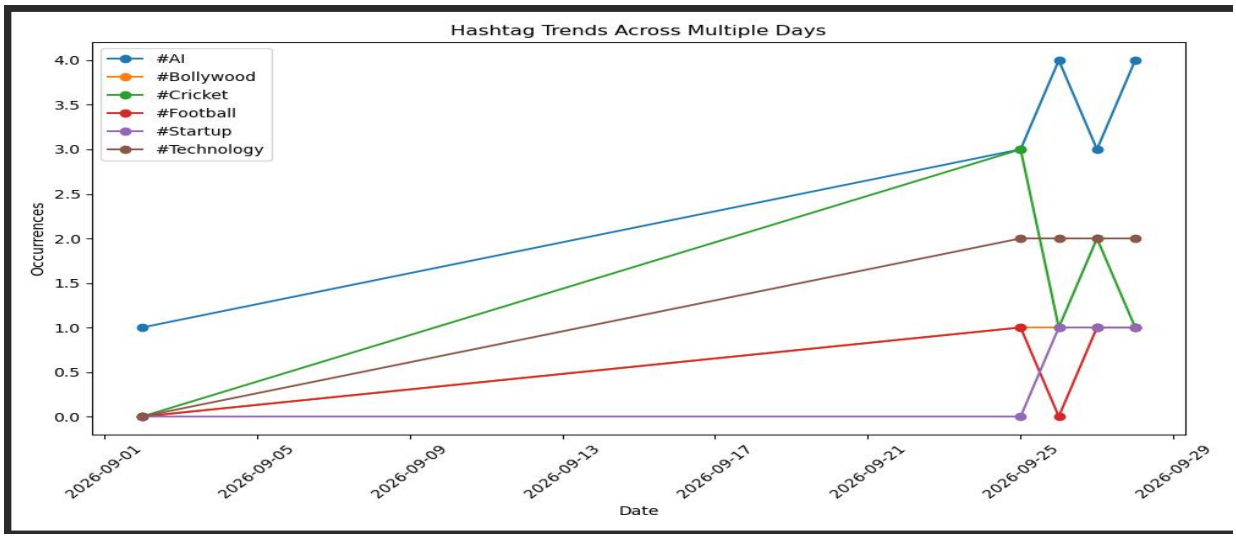

===== 3. HASHTAG POPULARITY BY REGION =====
+------+-----------+-----------+
|Region|Hashtag    |Occurrences|
+------+-----------+-----------+
|India |#AI        |13         |
|India |#Technology|8          |
|India |#Startup   |2          |
|India |#Cricket   |1          |
|UK    |#Bollywood |4          |
|USA   |#Cricket   |6          |
|USA   |#Football  |3          |
|USA   |#AI        |2          |
|USA   |#Startup   |1          |
+------+-----------+-----------+
```

```
+------+-----------+-----------+
|Region|Hashtag    |Occurrences|
+------+-----------+-----------+
|India |#AI        |13         |
|India |#Technology|8          |
|India |#Startup   |2          |
|India |#Cricket   |1          |
|UK    |#Bollywood |4          |
|USA   |#Cricket   |6          |
|USA   |#Football  |3          |
|USA   |#AI        |2          |
|USA   |#Startup   |1          |
+------+-----------+-----------+


================================================
4. HASHTAG CATEGORIES
================================================
+-------------+-----------+
|Category     |Occurrences|
+-------------+-----------+
|Technology   |23         |
|Sports       |10         |
|Entertainment|4          |
|Other        |3          |
+-------------+-----------+
```


Top 3 Trending Hashtags
Occurrences
0
2
4
6
8
10
12
14
#AI
#Technology
#Cricket
Hashtag


Hashtag Popularity by Category
Occurrences
0
5
10
15
20
Technology
Sports
Entertainment
Other
Category


Hashtag Trends Across Multiple Days
#AI
#Bollywood
#Cricket
#Football
#Startup
#Technology
Occurrences
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
2026-09-01
2026-09-05
2026-09-09
2026-09-13
2026-09-17
2026-09-21
2026-09-25
2026-09-29
Date

Tools / Technology used:

Apache Spark 3.x | PySpark | Python 3.11 | VS Code / Jupyter Notebook | Matplotlib (Optional Visualization)

Conclusion:
Thus, this practical successfully demonstrated the use of Apache Spark and PySpark for large-scale social media hashtag analysis. The dataset was loaded and validated using Spark DataFrames, after which hashtag frequencies were calculated and the most trending hashtags were identified. The analysis also compared hashtag trends across multiple days, studied regional popularity, and classified hashtags into Technology, Sports, and Entertainment categories. Three visualizations were generated to represent the major trends clearly. Overall, the practical shows how Spark-based analytics can help understand user interests, emerging trends, and popular topics, which can support better digital marketing strategies and business decision-making.

Github Repo:

https://github.com/tushar-2606/BDA-SHP-2026